Home Reading 10 — Optional PID Control

Preparation

Use this optional unit after Lectures 7–9. Allow about 25–30 minutes. It is separate from Project 2 and does not change Tutorial 1–8 submission credit.

Read selectively

Ogata, System Dynamics, 4th ed., printed pp. 501–505 (PDF pp. 512–516): controller actions. Focus on P, I, D and PI/PID; the electronic circuit is not needed here.

Lecture 10: slides 5–10 for the parallel PID convention; slides 14–16 for PI design and actuator demand; slides 19–21 for saturation and anti-windup. Use the slides to resolve a gap rather than repeat the whole textbook selection.

Check your understanding

1. For a constant nonzero error, which P, I or D contribution stays constant, grows, or becomes zero?

2. Why can a stable PI loop remove step offset? What feasibility condition must the actuator satisfy?

3. What is integrator windup? When should conditional integration stop, and when must it resume?

4. Why can derivative action cause a reference kick or amplify sensor noise? What does a filter change?

5. Does matching the desired denominator alone determine overshoot or guarantee a feasible controller? Name one further check.

Prepare for Tutorial 10

Try Q1. For Q2, write the first line of the PI characteristic equation and predict whether the heater will clip at startup. Explain your prediction before running MATLAB. Q3 is a short PID parameter audit. Extensions A–B are further practice.

You are ready when you can explain the terms above without copying definitions and show your own first step for Q2. Record one unresolved question if needed.

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